

Subject Description Form

Subject Code	AF4317
Subject Title	Derivative Securities
Credit Value	3
Level	4
Normal Duration	One Semester
Pre-requisite / Co-requisite/ Exclusion	Pre-requisite: AF3313 Business Finance
Objectives	The subject aims to provide students with the necessary knowledge, abilities and skills in relation to pricing and trading derivative securities and using derivative securities to manage risk in a corporation or financial institution.
Intended Learning Outcomes	Upon successful completion of the subject, students will be able to: - Evaluate the issues in pricing, hedging, and arbitrage in the derivative securities markets by applying tools in data analytics (BBA Outcome 14) - Evaluate the price determination of derivatives and derive the price of derivatives by using different models. (BBA Outcome 14) - Apply various trading and hedging strategies in real life situations to enhance portfolio returns. (BBA Outcome 14) - Evaluate the real-life situations in which financial derivatives resulted in success or failure while exploring everyday uses of financial derivatives. (BBA Outcome 14)
Subject Synopsis/ Indicative Syllabus	Introduction What derivatives are and their economic role. The organization of derivatives markets and how derivatives are traded. Recent development of derivatives in Hong Kong will be discussed. Futures and Forwards The structure of forward and futures markets. Principles of spot pricing. Principles of forward and futures pricing. Futures hedging strategies. Advanced futures strategies. Management Risk in Corporations Risk management function in a company. Avoiding derivatives losses. Options The structure of options markets. Principles of option pricing. Option pricing models. Basic option strategies. Advanced option strategies.
Teaching/Learning Methodology	Lectures will be conducted on the topics of the syllabus. Lecture time will be used flexibly for discussing key concepts and their applications with students and carrying out other learning activities with them. Such activities include group discussions, in-class exercises and student presentations of their work. To maximize benefits, students are encouraged to share their views and experiences actively with their lecturer and classmates.

Assessment Methods in Alignment with Intended Learning Outcomes	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed (Please tick as appropriate)					
			a	b	c	d		
	1. Group and Independent Assignments	20%	√	√	√	√		
	2. Mid-term Exam	25%	√	√				
	3. Final Exam	50%	√	√	√	√		
	4. Participation	5%	√	√	√	√		
	Total	100 %						
Student Study Effort Expected	Class contact:							
	▪ Lectures / Seminars					39 Hrs.		
	Other student study effort:							
	▪ Self-study					78 Hrs		
	Total student study effort					117 Hrs		
Reading List and References	<u>Required Text</u>							
	John Hull, <i>Fundamentals of Futures and Options Markets</i> , 9th Edition, Prentice Hall, 2016.							
	Supplementary Readings							
	Don Chance and Robert Brooks, <i>An Introduction to Derivatives and Risk Management</i> , 10th Edition, Cengage Learning, 2016.							
	John Marthinsen, <i>Risk Takers: Uses and Abuses of Financial Derivatives</i> , 2nd Edition, Pearson Addison Wesley, 2009.							
	Journals							
	<i>Journal of Future Markets</i> <i>Reviews of Derivatives Research</i>				<i>Derivatives Quarterly</i> <i>Financial Analysts Journal</i>			